



KENYA METHODIST UNIVERSITY

FACULTY OF SCIENCE AND TECHNOLOGY

DEPARTMENT OF PURE AND APPLIED SCIENCES

Course code : MATH 330

Course Title : OPERATIONS RESEARCH 1

Instructor : Kimathi Murungi

1.WHAT IS OPERATIONS RESEARCH?



Operations Research (O.R.), or operational research in the U.K, is a discipline that deals with the application of advanced analytical methods to help make better decisions. The terms management science and analytics are sometimes used as synonyms for operations research.

Employing techniques from other mathematical sciences, such as mathematical modeling, statistical analysis, and mathematical optimization, operations research arrives at optimal or near-optimal solutions to complex decision-making problems.

Operations research overlaps with other disciplines, notably industrial engineering and operations management. It is often concerned with determining a maximum (such as profit, performance, or yield) or minimum (such as loss, risk, or cost.)

Operations research encompasses a wide range of problem-solving techniques and methods applied in the pursuit of improved decision-making and efficiency, such as simulation, mathematical optimization, queuing theory, Markov decision processes, economic methods, data analysis, statistics, neural networks, expert systems, and decision analysis. Nearly all of these techniques involve the construction of mathematical models that attempt to describe the system.

Because of the computational and statistical nature of most of these fields, O.R. also has strong ties to computer science. Operations researchers faced with a new problem must determine which of these techniques are most appropriate given the nature of the system, the goals for improvement, and constraints on time and computing power.

The major sub-disciplines in modern operations research are:

- Computing and information technologies
- Environment, energy, and natural resources
- Financial Engineering
- Manufacturing, service science, and supply chain management
- Marketing Science
- Optimization
- Policy modeling and public sector work
- Revenue management
- Simulation
- Stochastic models
- Transportation.

History

As a formal discipline, operations research originated in the efforts of military planners during World War II. In the decades after the war, the techniques began to be applied more widely to problems in business, industry and society. Today, operations research is used by virtually every business and government throughout the world and remains an active area of academic research.

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Analytics

Operations Research is very closely related to Advanced Analytics.-_Analytics is defined as the scientific process of transforming data into insight for making better decisions.

What is Management Science?

Management science (MS), is an interdisciplinary branch of applied mathematics, engineering

and sciences that uses various scientific research-based principles, strategies, and analytical methods including mathematical modeling, statistics and algorithms to improve an organization's ability to enact rational and meaningful management decisions. The discipline is typically concerned with maximizing profit, assembly line performance, crop yield, bandwidth, etc or minimizing expenses, loss, risk, etc.

Overview

Management science is concerned with a number of different areas of study including developing and applying models and concepts that may prove useful in helping to illuminate management issues and solve problems. Management science research can be done on three levels:

- A fundamental level that lies in three mathematical disciplines: probability, optimization, and dynamic systems theory,
- A modeling level that builds models, gathers data, and analyzes them mathematically, and
- An application level, just as any other engineering discipline that has strong aspirations to make a practical impact in the real world.

Theory

Some of the fields that are included in Management Science are:

Data mining

Optimization

Decision analysis

Probability and statistics

Engineering

Project management

Forecasting

Simulation

Game theory

Social network

Industrial engineering

Transportation forecasting models



Logistics

Supply chain management

Mathematical modeling

as well as many others.

Applications

Applications of management science are abundant in industry such as airlines, manufacturing companies, service organizations, military branches, and in government. The range of problems and issues to which management science has contributed insights and solutions is vast. It includes:

- scheduling airlines, both planes and crew,
- deciding the appropriate place to place new facilities such as a warehouse or factory,
- managing the flow of water from reservoirs,
- identifying possible future development paths for parts of the telecommunications industry,
- establishing the information needs and appropriate systems to supply them within the health service, and
- identifying and understanding the strategies adopted by companies for their information systems.

2.LINEAR PROGRAMMING (LP)

The word „linear“ means that the relationships handled are those represented by straight lines i.e. the relationships are of the form $y = a + bx$ and the word “programming” means taking decisions systematically. Thus linear programming is a decision making technique under given constraint on the assumptions that the relationships among the variables representing different phenomena are linear.

Linear programming is generally used in solving maximization (Sales or Profit maximization or minimization (Cost minimization) problems subject to certain assumptions.

Linear programming is the maximization (or minimization) of a linear function of variables subject to a constraint of linear inequalities.

Linear programming is a mathematical technique designed to assist the organization in optimally allocating its available resources under conditions of certainty in problems of scheduling, product mix and so on.

In Linear programming, we start by formulating the objective function (Function that is either going to be maximized or minimized e.g. profit and cost) and the set of constraints/limitations.

FIELDS WHERE LINEAR PROGRAMMING CAN BE APPLIED

- 1) Agricultural applications- LP can be applied in farm management problems so far it relates the allocation of resources such as acreage, labour water supply, or working capital in such a way to maximize net revenue.
- 2) Contract Awards- LP can be used to evaluate tenders to guarantee that tenders are made in the cheapest way.
- 3) Industrial Application – Transportation problems concerning cost minimization can be solved by this technique. The Technique can as well be adopted in solving the problems of production (product mix) and inventory control.

Thus, linear programming is the most widely used technique of decision making in business and industry in modern times in various fields.

ASSUMPTIONS IN LP MODELS

A Linear programming model is based on the following assumptions:

1) Proportionality:

A basic assumption of LP is that proportionality exists in the objective function and the constraints. This assumption implies that if a product yields a profit of £10, the profits earned from that sale is 12 such products will be £(10x12)= £120. This may not always be the case because of quantity discounts. Further, even if the sale price is constant, the manufacturing cost may vary with the number of units produced and so may vary the profit per unit. Likewise, it is assumed that if one product requires processing time of 5 hours, then ten such products will require processing time of $5 \times 10 = 50$ hrs. This may also not be true as the processing time per unit often decreases with increase in the number of units produced. The real situations may not be strictly linear. However, assumed linearity represents their close approximation and provides very useful answers.

2) Additivity

It means that if we use t_1 hours on machine A and t_2 hours to make products 2, then the total time required to make products 1 and 2 on machine A is $t_1 + t_2$ hours. This however, is true only if the change-over time from product 1 to product 2 is negligible. Some products may not behave in this way e.g. when several liquids of different chemical compositions are mixed; the resulting volume may not be equal to the sum of the volumes of the individual liquids.

3) Continuity

It is assumed that decision variables are continuous i.e., they are permitted to take any non-negative values that satisfy the constraints. However, there are problems whereby variables are restricted to have integral values only.

4) Certainty

LP makes no allowance for uncertainty in the estimate made in other words; the certainty assumption is that all the parameters of the model are known constraints. In real problems, this assumption is seldom satisfied precisely. LP models usually are formulated in order to select some future course of action.

5) Divisibility

Quantities are infinitely divisible i.e., any fractional or decimal answer is valid. This means allowing production of 14.5 chairs for example. The figure can then be rounded off to the next lower figure. For problems requiring integer solutions, we have integer linear programming.

6) **Non –Negativity of the decision variables**- only positive values are considered.

7) **Time factor is ignored**- all production is assumed to be instantaneous.

Advantages of LP

- 1) It helps in attaining optimum use of productive factors. More efficient use of manpower and machines can be obtained by use of LP.
- 2) It improves the quality of decisions. A manager who makes use of LP becomes more objective than subjective.
- 3) It highlights the bottlenecks in the production process. When bottlenecks occur, some machines cannot meet demand while others remain idle for at least part time.
- 4) It can go a long way in improving the knowledge and skill of tomorrow's managers.

Limitations of Linear Programming

- 1) It treats all relationships as linear e.g. if direct cost of producing 10 units is Ksh.100, then 10 units will be produced at Ksh.200. In practice, however the direct costs of production may increase as output approaches capacity. It is assumed that the objective function and constraints can be represented as linear inequalities or equations. This will not always be the case.
 - 2) It is assumed that any activity is infinitely divisible. Evidently in many cases such assumption would be physically unjustifiable.
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- 3) All the parameters in linear programming model are assumed to be known constants. Whereas, in reality they are frequently neither known nor constants.
- 4) It takes into account single objective only i.e. profit maximization, cost minimization etc. However, many organizations keep in view multiple objectives such as capacity utilization, profit maximization, limiting overtime etc. in arriving at strategic management decision.
- 5) It can be adopted only under the conditions of certainty i.e. when the quantity of resources available, per unit contribution, cost, etc. are known with certainty.

METHODS/TECHNIQUES OF SOLVING LP PROBLEMS

- 1) General allocation method.
- 2) Transportation method.
- 3) Assignment method.

General Allocation methods

The Technique formulates a mathematical model which portrays the interaction and behaviour of the essential elements of a situation in the form of a set of algebraic expressions. The expressions are either equations or inequalities. The inequalities are converted to equations for both the plotting of graphs and the formulation of matrices. In a problem consisting of only two variables or unknowns, a graphical method may be used to obtain the solution. In case of 3 or more variables we use “Simplex method”.

General allocation LP problems can be solved by the help of the following methods:

- i) Graphical methods
- ii) Analytical method(Trial & Error method)
- iii) Simplex method.

The starting point in solving a Linear programming problem is to formulate a LP model. The steps involved in formulating a LP model are:

- 1) Identify the decision variables to be determined and express them in term of algebraic symbols.
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- 2) Identify all the limitations or constraints in the given problem and then express them as linear inequalities or equations in terms of the above defined decision variables.

Mathematical formulation of LP problem

If there are „n“ decision variables and m constraints in the problem, the mathematical formulation of the LP is.

Optimize (maximize or minimize) $Z = C_1X_1 + C_2X_2 + \dots + C_nX_n$

Subject to the Constraints

$$\begin{aligned}
 a_{11}X_1 + a_{12}X_2 + \dots + a_{1n}X_n &\leq b_1 \\
 &\geq \\
 a_{21}X_1 + a_{22}X_2 + \dots + a_{2n}X_n &\leq b_2 \\
 &\geq \\
 &\text{"} \\
 &\text{"} \\
 &\dots \\
 a_{m1}X_1 + a_{m2}X_2 + \dots + a_{mn}X_n &\leq b_m \\
 &\geq
 \end{aligned}$$

GRAPHICAL METHOD

The method is appropriate if the decision maker has two decision variables

Steps

- 1) Formulate the LP problem.
- 2) Graph the constraint inequalities as follows: Treat each inequality as though it were equality and for each equation select two sets of points arbitrary. Plot each set of points and connect them with appropriate lines.
- 3) Identify the solution space or feasible region which satisfies all the constraints simultaneously.
- 4) Locate the solution points on the feasible region. These points always occur at the corner points of the feasible region.
- 5) Evaluate the objective function at each of the corner points of the feasible region.
- 6) Identify the optimum value of the objective function.

Illustration

The LP model for a given problem is

$$\text{Maximize } Z = 3X_1 + 4X_2$$

$$\text{Subject to } 5X_1 + 4X_2 \leq 200$$

$$3X_1 + 5X_2 \leq 150$$

$$5X_1 + 4X_2 \geq 100$$

$$8X_1 + 4X_2 \geq 80$$

$$X_1, X_2 \geq 0$$

The solution space satisfying the given constraints and meeting the non-negativity restrictions

$X_1 \geq 0$, and $X_2 \geq 0$ is shown in the figure below.

$$5X_1 + 4X_2 = 200$$

X_1	0	40
X_2	50	0

$$3X_1 + 5X_2 = 150$$

X_1	0	50
X_2	30	0

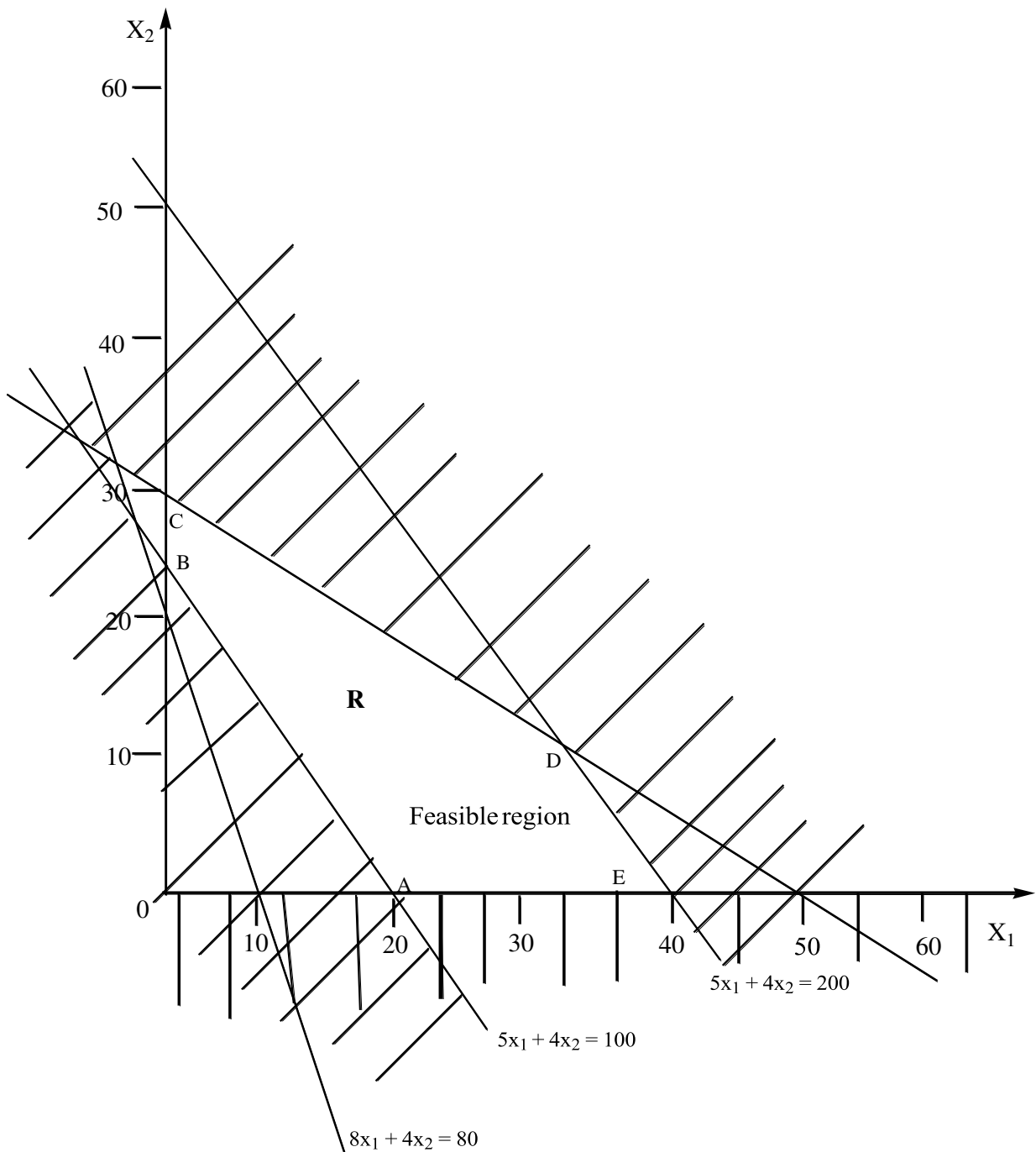
$$5X_1 + 4X_2 = 100$$

X_1	0	20
X_2	25	0

$$8X_1 + 4X_2 = 80$$

X_1	0	10
X_2	20	0





Any point in the unshaded region is a feasible solution to the given problem. The coordinates of the vertices of this convex region ABCDE are A (20,0), B (0,25), C (0,30), D (400/13, 150/13)= (31,12) and E (40,0).

Values of the objective function $Z = 3X_1 + 4X_2$ at these vertices are:

$$Z(A) = 60, Z(B) = 100, Z(C) = 120, Z(D) = 1800/13, Z(E) = 120$$

Since the maximum value of Z is $1800/13$ which occurs at the vertex $D (400/13, 150/13)$ the solution to the given problem is

$$X_1, 400/13, X_2, 150/13 \text{ and } Z \text{ max} = \text{£ } 1800/13$$

$$= \text{£ } 138.46$$

Illustration

The final product of a firm has a requirement that it must weigh exactly 150kg. The two raw-materials used in the manufacture of this product are A with a cost of £ 2 per unit and B with a cost of £ 8 per unit. At least 14 units of B and no more than 20 units of A must be used. Each unit of A weighs 5kg and each unit of B weighs 10 kg.

How much of each type of raw material should be used for each unit of the final product if cost is to be maximized.

Solution

We write the problem in form of a LP model

$$\text{Minimize } Z = 2X_1 + 8X_2 \text{ (objective function)}$$

Where Z = cost of the final product

X_1 = number of units used of raw material A

X_2 = number of units used of raw material B

Subject to:

$$\underline{5X_1 + 10X_2}$$

$$X_2 \geq 14$$

$$X_1 \leq 20$$

$$X_1 \geq 0 \} \text{ Non – negativity condition}$$

$$5X_1 + 10X_2 = 150$$

Now, present the model graphically



Self assessment questions

Question one

A firm engaged in producing two models viz model X_1 and model X_2 performs only three operations – painting, assembly and testing. The relevant data are as follows:

	Hours required for each unit.		
Unit sale price	Assembly	Painting	Testing
Model X_1 £ 50	1.0	0.2	0.0
Model X_2 £ 80	1.5	0.2	0.1

Total number of hours available each week is as under:

Assembly 600

Painting 100

Testing 30

The firm wishes to determine its weekly product mix so as to maximize revenue

Required:

Write up the model and then solve the product mix problem.

Question two

XYZ Company Ltd has two bottling plants, one located at Voi and another at Mombasa. Each plant produce three drinks, A, B & C. The numbers of bottles produced per day are as follows:

PLANT AT

	Mombasa	Voi
Drink	(M)	(V)
A	1,500	1,500
B	3,000	1,000
C	2,000	5,000

A market survey indicates that during the month of December, there will be a demand of 200,000 bottles of A, 40,000 Bottles of B and 44,000 bottles of drink C. The operating costs per day for plants at Mombasa & Voi are 600 and 400 monetary units. For how many days should each plant be run in December so as to minimize the production cost, while still meeting the market demand?

ANALYTICAL METHOD (TRIAL & ERROR) METHOD OF SOLVING LP PROBLEMS

Graphical method cannot be applied to solve LP problems involving more than 2 variables. In such cases, analytical method (Also called trial and error method) is quite helpful.

This method also forms a good basis to grasp the more powerful simplex method. How the method works is demonstrated by the illustration below

Example

Linear programming model for a problem is:

$$\text{Maximize } Z = 3X_1 + 4X_2$$

$$\text{Subject to } X_1 + X_2 \leq 450$$

$$2X_1 + X_2 \leq 600$$

$$\text{Where } X_1, X_2 \geq 0$$

The problem is first expressed in standard form by the introduction of slack variables S_1 and S_2 as shown below:

$$\text{Maximize } Z = 3X_1 + 4X_2 + 0S_1 + 0S_2$$

$$\text{Subject to } X_1 + X_2 + S_1 = 450$$

$$2X_1 + X_2 + S_1 = 600$$

$$X_1, X_2, S_1, S_2 \geq 0$$

The slack variables represent dummy (fictitious or imaginary) products with Zero contribution per unit. Here total number of variables, $M+N = 4$ and number of constraints, $m=2$. A basic solution can be obtained by setting any of the n (2) variables equal to Zero and then solving the constraints equations. The total number of basic solutions will be

$$\frac{n!}{m!} = 6$$

The table below gives a summary of the characteristics of the various basic solutions.

S. No. of the basic solution	Basic variables	Non basic variables	Values of the basic variables given by constraints equations	Value of the objective functions	Is the solution feasible are all $X_i \geq 0$?	Is the solution non degenerate are all basic variables > 0 ?	Is the solution feasible and optimal?
1	X_1, X_2	S_1, S_2	$X_1 + X_2 = 450$ $2X_1 + X_2 = 600$ $X_1 = 150$ $X_2 = 300$	1,650	Yes	Yes	No
2	X_1, S_1	X_2, S_2	$X_1 + S_1 = 450$ $2X_1 = 600$ $X_1 = 300$ $S_1 = 150$	900	Yes	Yes	No
3	X_1, S_2	X_2, S_1	$X_1 = 450$ $2X_1 + S_2 = 600$ $X_1 = 450$ $S_2 = -300$	-	No	No	No
4	X_2, S_1	X_1, S_2	$X_2 + S_1 = 450$	-	No	No	No

			$X_2 = 600$ $X_1 = -150$				
5	X_2, S_2	X_1, S_1	$X_2 = 450$ $X_2 + S_2 = 600$ $S_2 = 150$	1,800	Yes	Yes	Yes
6	S_1, S_2	X_1, X_2	$S_1 = 450$ $S_2 = 600$	0	Yes	Yes	No

Out of these solutions, solutions in which all basic variables X_1 are ≥ 0 will be basic feasible solutions in which all basic variables are > 0 will be non-degenerate basic feasible and the basic feasible solution that optimizes the objective function will be the optimal basic feasible solution.

It may be seen that solutions 1,2,5 and 6 are basic feasible; they are also non-degenerate out of these solution 5 gives the maximum value of the objective function Z and is therefore, the optimal solution. Solutions 3 and 4 are infeasible and are to be discarded from consideration. Thus the optimal solution to the problem is.

$$X_1 = 0, X_2 = 450;$$

$$Z_{\max} = 1,800$$



$$\text{Maximize } Z = 12X_1 + 8X_2 + 14X_3 + 10X_4$$

$$\text{Subject to } 5X_1 + 4X_2 + 2X_3 + X_4 = 100$$

$$2X_1 + 3X_2 + 8X_3 + X_4 = 75$$

$$\text{Where } X_1, X_2, X_3, X_4, \text{ are all } \geq 0$$

The answer is

$$X_1 = 25/3, X_2 = 0$$

$$X_3 = 0, X_4 = 175/3$$

$$\text{And } Z_{\max} = 2050/3$$

SIMPLEX METHOD FOR MAXIMIZING

Is a step by step arithmetic method of solving LP problems whereby one moves progressively from a position of, say zero production, and therefore zero contribution until no further contribution can be made. Each step produces a feasible solution and each step produces an answer better than the one before i.e. either greater contribution in maximizing problems, or less cost in minimizing problems.

The mathematics behind Simplex method are complex and therefore we are not going to explain why the method works but we describe how to use the technique.

FORMULATING THE SIMPLEX METHOD

We state the problem in the standardized manner. i.e. objective for and a number of constraints which are inequalities either of $\geq$ or $\leq$ type.

The inequalities are converted into equations by adding an extra variable called a Slack Variable. Slack variables represent any unused capacity in the constraint

NB:

- 1) Each constraint will have its own slack variables.
- 2) Once the slack variables have been incorporated into the constraint the simplex method automatically assigns it an appropriate value at each iteration.

Maximizing Examples

$$\begin{aligned}\text{Maximize} \quad & 8 X_1 + 5 X_2 + 10 X_3 \\ \text{Subject to} \quad & 2 X_1 + 3 X_2 + X_3 \leq 400 \\ & X_1 + X_3 \leq 150 \\ & 2 X_1 + 4 X_3 \leq 200 \\ & X_2 \leq 50 \\ & X_1 \geq 0, X_2 \geq 0, X_3 \geq 0\end{aligned}$$

Solution

We make the inequalities in the constraints into equalities by adding a “slack variable” in each constraint, thus;

$$\begin{aligned}\text{Maximize:} \quad & 8 X_1 + 5 X_2 + 10 X_3 \\ \text{Subject to:} \quad & 2 X_1 + 3 X_2 + X_3 + X_4 = 400 \\ & X_1 + X_3 + X_5 = 150\end{aligned}$$

$$2 X_1 + 4 X_3 + X_6 =$$

$$200 X_2 + X_7 = 50$$

NB: X_4 , X_5 , X_6 & X_7 are the slack variables and represent the spare capacity in the limitations

We set the initial simplex tableau by arranging the objective function and equalized constraints.

Solution Variable	Products			Slack Variables				Solution quantity
	X_1	X_2	X_3	X_4	X_5	X_6	X_7	
X_4	2	3	1	1	0	0	0	400
X_5	1	0	1	0	1	0	0	150
X_6	2	0	4	0	0	1	0	200
X_7	0	1	0	0	0	0	1	50
Z	8	5	10	0	0	0	0	0

Note:

Values in the body of the table are the values from the objective function and constraints.

The variable „Z“ has been used for the objective function and represents total contribution.

The table shows that $X_4 = 400$, $X_5 = 150$, $X_6 = 200$, $X_7 = 50$ and $Z = 0$

The tableau shows a feasible solution, that of nil production, nil contribution, and maximum un used capacity as represented by the values of the Slack variables X_4, X_5, X_6 , & X_7

Although feasible, this plan can obviously be improved as follows:

Step 3

Improve the previous feasible solution by making as many as possible of the product with the most contribution, i.e. the highest figure in the Z-row

The number that can be made will be limited by one or more of the constraints becoming operative thus:

Select highest contribution in Z row i.e. 10 under X_3 .

Divide the positive number in the X_3 column into the solution quantity column

$$400 \div 1 = 400$$

$$150 \div 1 = 150$$

$$200 \div 4 = 50$$

$$50 \div 0 = \text{ignore}$$

Select the row that gives the lowest answer (in this case the row identified is X_6 . Ring the element which appears in both identified column (X_3) and identified row (X_6) this element is called the pivot element thus,

Initial Simplex tableau reproduced:

Table 2

Solution Variable	Products			Slack Variables				Solution quantity
	X_1	X_2	X_3	X_4	X_5	X_6	X_7	
X_4	2	3	1	1	0	0	0	400
X_5	1	0	1	0	1	0	0	150
X_6	2	0	4	0	0	1	0	200
X_7	0	1	0	0	0	0	1	50
Z	8	5	10	0	0	0	0	0

Pivot element

Divide all elements in the identified row (X_6) by the value of pivot element (4) and change the solution variable to the leading of the identified column (X_3) thus

OLD ROW

X_6	2	0	4	0	0	1	0	200
-------	---	---	---	---	---	---	---	-----

New row

X_3 $\frac{1}{2}$ 0 1 0 $0\frac{1}{4}$ 0 50

Enter this row in a new tableau

Second Simplex Tableau: Table 3

Rows No.	Solution Variable	Products			Slack Variables				Solution quantity
		X_1	X_2	X_3	X_4	X_5	X_6	X_7	
1	X_4	2	3	1	1	0	0	0	400
2	X_5	1	0	1	0	1	0	0	150
3	X_6	2	0	4	0	0	1	0	200
4	X_7	0	1	0	0	0	0	1	50
5	Z	8	5	10	0	0	0	0	0

Notes

- The row numbers have been included to aid the explanatory material which follows and from no. part of the Simplex tableau.

b) It will be seen that this second tableau is identical to the first except for row 3 which was calculated above

c) Row 3 means that 50 units of X_3 are to be produced

As a consequence of producing 50 units of X_3 is necessary to adjust the other rows so as to take up the appropriate number of hours, components etc and to show the contribution produced for its 50 units of X_3 . This is done by repetitive row by row operations using row 3 which makes all other elements in the pivot elements column into zeros.

To maintain each row as an equality, we alter each element along the row on both sides on the equality sign. Using row one in second tableau as an example the process is as follows.

Row 1	X_4	2	3	1	1	0	0	0	400
Minus Row 3	X_3	$\frac{1}{2}$	0	1	0	0	$\frac{1}{4}$	0	50
Produces a new row	X_4	$1\frac{1}{2}$	3	0*	1	0	$-\frac{1}{4}$	0	350

Notes:

- This new row will be inserted into a third tableau along with all other altered rows and row three from the second tableau.
- The aim of this row operation was to produce the zero (marked *). In this case a simple subtraction was all that was necessary but to make a zero in other cases may require further operations using row e as a basis.

The other rows are operated on a similar fashion.

Row 2	X_5	1	0	1	0	1	0	0	150
Minus Row 3	X_3	$\frac{1}{2}$	0	1	0	0	$\frac{1}{4}$	0	50
Produces a new row	X_5	$\frac{1}{2}$	3	0*	1	0	$-\frac{1}{4}$	0	100

Row 4 needs no operation because the element in column X_3 is already zero

Row 5	Z	8	5	10	0	0	0	0	0
Minus 10 x Row 3	X_3	5	0	10	0	0	$2\frac{1}{2}$	0	500
Produces a new row	Z	3	5	0*	0	0	$-2\frac{1}{2}$	0	-500

Note:

- To produce the required zero (0*) it was necessary to multiply the row 3 by 10 and then subtract from row 5
- The “-500” at end of new Z row is the contribution earned by 50 units of X_3 at £ 10 i.e. £500. The negative sign is merely a result of simplex method and the fact that the contribution is shown as a negative figure can be disregarded.

Now the new third tableau can be produced thus:

Rows No.	Solution Variable	Products			Slack Variables				Solution quantity
		X ₁	X ₂	X ₃	X ₄	X ₅	X ₆	X ₇	
6 (Row 1 –Row 3)	X ₄	1 1/2	3	0	1	0	- 1/4	0	350
7 (Row 2-Row 3)	X ₅	1/2	0	0	0	1	- 1/4	0	100
8(i.e Row 3)	X ₃	1/2	0	1	0	0	1/4	0	50
9 (i.e row 4)	X ₇	0 ²	1	0	0	0	0 ⁴	1	50
10 (Row 5 – 10 x Row 3)	Z	3	5	0	0	0	- 2 1/2	0	-500

To produce subsequent tableaux and eventually an optimum solution, steps 4 to 7 are repeated until no positive numbers can be found in Z row.

From row 10, the maximum contribution is 5

Therefore, X₂ column is chosen

Positive numbers in X₂ column are divided into the solution quantitative and the lowest result is selected.

i.e. Row 6 $350 \div 3 = 116 \frac{2}{3}$

Row 7 $100 \div 0 = \text{Ignore}$

Row 8 $50 \div 0 = \text{Ignore}$

Row 9 $50 \div 1 = 50$

Row 9 is selected & pivot element identified and solution variable altered to X₂ thus.

Row 9 X₂ therefore, 0 1 0 0 0 0 1 =50

The pivot element is already 1 and hence no further action is necessary on it but other elements in the pivot element column (X₂) must be made into zeros by using row operations based on row 9, thus

Row 6	X ₄	1 1/2	3	0	1	0	-1/4	0	350
Minus 3x Row 9	X ₂	0	3	0	0	0	0	3	150
Produces a new row	X ₄	1 1/2	0*	0	1	0	-1/4	-3	200

We multiplied row 9 by 3 to produce zero in column X₂ (0*) . Row 7 and 8 need no operates because elements in column X₂ are already zero

Row 10	Z	3	5	0	0	0	-2 1/2	0	-500
Minus 5x Row 9	X ₂	0	5	0	0	0	0	5	250
Produces a new row	Z	3	0*	0	0	0	-2 1/2	-5	-750

We enter this in a new tableau (fourth)

Fourth Simplex tableau; Table 5

Rows No.	Solution Variable	Products			Slack Variables				Solution quantity
		X ₁	X ₂	X ₃	X ₄	X ₅	X ₆	X ₇	
11 (Row 6 – 3 x Row 9)	X ₄	1 1/2	0	0	1	0	- 1/4	-3	200
12 (as Row 7)	X ₅	1/2	0	0	0	1	- 1/4	0	100
13 (i.e Row 8)	X ₃	1/2	0	1	0	0	1/4	0	50
14 (pivot row as row 9)	X ₂	0	1	0	0	0	0	1	50
15(Row 10 – 5 Row 3)		3	0	0	0	0	- 2 1/2	-5	-750

The tableau shows that 50 units of products B and C could be made (X₂& X₃ = 50) and contribution of £ 750 would be made (Z=-750)

Because there is still a positive number in Z row, the procedure is repeated in precisely the same manner. Column X₁ is chosen and divided into solution quantities and lowest number selected.

Row 11 $200 \div 1 \frac{1}{2} = 133 \frac{1}{3}$

Row 12 $100 \div \frac{1}{2} = 200$

Row 13 $50 \div \frac{1}{2} = 100$

Row 14 $50 \div 0 = \text{Ignore}$

Therefore, row 13 is selected and pivot element identified and the solution variable altered to X₁ thus.

Row 13 X₁ 1/2 0 1 0 0 1/4 0 =50

Pivot element must be made into a 1 so the whole row is multiplied by 2 thus

Row 13 X₁ 1 0 2 0 0 1/2 0 =100

The rest of the elements in column X₁ must be made into Zeros by the usual row operation.

Row 11	X ₄	1 1/2	0	0	1	0	- 1/4	-3	200
Minus 1 1/2 x Row 13	X ₁	1 1/2	0	3	0	0	3/4	0	150
Produces a new row	X ₄	0*	0	-3	0	0	-1	-3	50

Row 12	X ₅	1/2	0	0	0	1	- 1/4	0	100
Minus 1/2 x Row 13	X ₁	1/2	0	1	0	0	1/4	0	50
Produces a new row	X ₅	0*	0	-1	0	1	-1/2	0	50

Row 14 needs no operation because the element in column X_1 is already zero

Row 15	Z	3	0	0	0	0	$-2\frac{1}{2}$	-5	-750
Minus 3 x Row 13	X_1	3	0	6	0	0	$-1\frac{1}{2}$	0	300
Produces a new row	Z	0*	0	-6	0	0	$^{-4}$	-5	-1,050

The new rows produced can now be entered into a 5th tableau thus:

Fifth simplex tableau: Table 6

Rows No.	Solution Variable	Products			Slack Variables				Solution quantity
		X_1	X_2	X_3	X_4	X_5	X_6	X_7	
16 (Row 11 $-1\frac{1}{2}$ x Row 18)	X_4	0	0	-3	1	0	-1	-3	50
17 (Row 12 $-1\frac{1}{2}$ x Row 18)	X_5	0	0	-1	0	1	$-1\frac{1}{2}$	0	50
18 (Pivot Row 2 x Row 13)	X_1	1	0	2	0	0	$\frac{1}{2}$	0	100
19 (as row 14) X_2		0	1	0	0	0	0	1	50
20 (Row 15 -3 Row 13) Z		0	0	-6	0	0	-4	-5	-1,050

There are no positive values in the Z row and therefore optimum solution has been reached.

Optimum product mix:

$X_1 = 100$ i.e. produce 100 units of product A

$x_2 = 50$ i.e. produce 50 units of product B

Value of slack variables

$X_4 = 50$ i.e. there are 50 machines hours unused at optimum

$X_5 = 50$ i.e. there are 50 components unused at optimum.

The other two slack variables X_6 & X_7 do not have values in the Solution Quantity column.

Their values are both zero which means X_6 & X_7 have no unused capacity at optimum and the constraints they represent are fully utilized.

Contribution & resource valuations

Z has a solution quantity of -1,050. This means at optimum, the max contribution is £ 1,050 (this can be confirmed by calculating the contributions of the optimum product mix i.e.

$$100A \text{ and } 50 B = 100 (8) + 50 (5) = 1,050.$$

The value -6 under product X_3 , the product that was not in the optimum plan, means that if any unit of X_3 was produced then overall contribution would fall by £6.

The values of Row 20 for the slack variables for the slack variables are of great importance. These are the valuations of resources are known as shadow prices. These have the following meanings:

$X_4 = 0$ i.e. there are no value to be gained from increasing machine hours.

$X_5 = 0$ i.e. no value to be gained from increasing number of components.

$X_6 = -4$ i.e. for every extra kilo of alloy available £4 extra overall contribution would be gained.

$X_7 = -5$ i.e. for every extra unit of B that was allowed to be produced overall contribution would increase by £5

It will be seen that only the constraints X_6 & X_7 that are binding, i.e. are fully utilized, have non-zero shadow prices. This is a general rule, for there would be no value in increasing the availability of a resource already in surplus.

The shadow prices of the binding constraints can be used to confirm again the overall contribution thus:

Alloy availability $200\text{kg} \times £4 = £800$

Sales constraints $50 \text{ units} \times £5 = \underline{£250}$

£1,050



Self Assessment Questions

A firm manufactures three products A, B, and C. The profits per unit product are £3, £2 and £4 respectively. The firm has two machines and the required processing time in minutes for each machine on each product is given below.

	Product		
	A	B	C
Machine X	4	3	5
Y	2	2	4

Machines X and Y have 2,000 and 1,500 machine minutes respectively. The firm must manufacture 11A's, 200 B's and 50 C's but no more than 150 A's. Set up an L.P model to maximize the profit and solve it.

Answer

$$\text{Maximize } Z = 3X_A + 2X_B + 4X_C$$

$$\text{Subject to } 4X_A + 3X_B + 5X_C \leq 2,000$$

$$2X_A + 2X_B + 4X_C \leq 1,500$$

$$X_A \geq 100$$

$$X_A \leq 150$$

$$X_B \geq 200$$

$$X_C \geq 150$$



3. TRANSPORTATION AND ASSIGNMENT MODELS

Introduction

Transportation models deal with problems concerning as to what happens to effectiveness function when we associate each of a number of origins (sources) with each of a possibly different number of destinations (jobs).

The total movement from each origin and total movement to each destination is given and it is desired to find how the associations should be made subject to the limitations on totals. In such problems, sources can be divided among the jobs or jobs may be done with a combination of sources. The distinct feature of transportation problems is that sources and jobs must be expressed in terms of only one kind of unit.

Suppose that there are „m“ sources and „n“ destinations. Let a_i be the number of supply units available at source i ($i = 1, 2, 3,$

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 — — —
 — n)
 and
 let
 b_j be
 the
 num
 ber
 of

demand units required at destination j ($j = 1, 2, 3, \dots, n$)

Let C_{ij} represent the unit transportation cost for transporting the units from source i to destination j . The objective is to determine the number of units to be transported from source i to destination j so that the total transportation cost is minimum. In addition, the supply limits at the sources and the demand requirements at the destinations must be satisfied exactly.

If X_{ij} ($X_{ij} \geq 0$) is the number of units shipped from source i to destination j , then the equivalent linear programming model will be:

Find X_{ij} ($i = 1, 2, 3, \dots, m, j = 1, 2, 3, \dots, n$)

In order to

Minimize
$$Z = \sum_{i=1}^m \sum_{j=1}^n C_{ij} X_{ij}$$

Subject to

$$\sum_{i=1}^m a_i = \sum_{j=1}^n b_j$$

$$\sum_{j=1}^n X_{ij} = a_i, i = 1, 2, 3, \dots, m,$$

And

$$\sum_{i=1}^m X_{ij} = b_j, j = 1, 2, 3, \dots, n,$$

Where $X_{ij} \geq 0$

The two sets of constraints will be consistent i.e. the system will be balance if

$$\sum_{i=1}^m a_i = \sum_{j=1}^n b_j$$

Equality sign of the constraints causes one of the constraints to be redundant (and hence- it can be deleted) so that the problem will have (m+n-1) constraints and (m x n) unknown.

Note that a transportation problem will have a feasible solution only if the above restriction is satisfied. Thus, $\sum_{i=1}^m a_i = \sum_{j=1}^n b_j$ is necessary as well as a sufficient condition for transportation problem to have a feasible solution. Problems that satisfy this condition are called Balanced Transportation problems.

Techniques have been developed for solving balanced or standard transportation problems only. It follows that any non-standard problem in which the supplies and demands do not balance, must be converted to a standard transportation problem before it can be solved. This conversion can be achieved by the use of a dummy source/destination.

The above information can be put in the form of a general matrix shown below:

		1	2	3	-----j-----	n	Supply
Sources or Origins	1	$C_{11} X_{11}$	$C_{12} X_{12}$	$C_{13} X_{13}$	$C_{1j} X_{1j}$	$C_{1n} X_{1n}$	a_1
		$C_{21} X_{21}$	$C_{22} X_{22}$	$C_{23} X_{23}$	$C_{2j} X_{2j}$	$C_{2n} X_{2n}$	a_2
	2	$C_{31} X_{31}$	$C_{32} X_{32}$	$C_{33} X_{33}$	$C_{3i} X_{3i}$	$C_{3n} X_{3n}$	a_3
		.	.	.	.	.	
	3	.	.	.	.	.	
.		$C_{i1} X_{i1}$	$C_{i2} X_{i2}$	$C_{i3} X_{i3}$	$C_{ij} X_{ij}$	$C_{in} X_{in}$	a_i

	$C_{m1} X_{m1}$	$C_{m2} X_{m2}$	$C_{m3} X_{m3}$	$C_{mj} X_{mj}$	$C_{mn} X_{mn}$	a_n
i						
m						
Demand	b_1	b_2	b_3	b_j	b_n	

In the table C_{ij} , $i = 1, 2, \dots, m$; $j = 1, 2, \dots, n$,
 is the unit shipping cost from the i^{th} origin to the j^{th} destination. a_i is the supply available at origin i and b_j is the demand at destination j .

Definitions

1. Feasible Solution

A feasible solution to a transportation problem is a set of non-negative allocations, X_{ij} that satisfied the rim (row and column) restrictions.

2. Basic Feasible Solution

Solution that contains no more than $m+n-1$ non-negative allocations, where m is the number of rows and n is the number of columns of the transportation problem.

3. Optimal Solution

A feasible solution (not necessarily basic) that minimizes (maximizes) the transportation cost (profit) is called an optimal solution.

4. Non-degenerate Basic Feasible Solution

A basic feasible solution to a $(m \times n)$ transportation problem is said to be non-degenerate if,

- a) The total number of non-negative allocation is exactly $m+n-1$ (number of independent constraints equations), and
- b) These $m+n-1$ allocations are in independent positions.

5. Degenerate Basic Feasible Solution

A basic feasible solution which the total number of non-negative allocations is less than $m+n-1$ is called degenerate basic feasible solution

Methods of solving transportation problems

1. Northwest corner method (NWCN)
2. Row minima method
3. Column minima method
4. Least – cost method (or matrix minima method or lowest cost entry method)
5. Vogel’s approximation method (VAM) or penalty method or Regret method.

In this lecture, we discuss the VAM

VAM

VAM is a heuristic method and is preferred to the methods listed above. In the transportation matrix if an allocation is made in the second lowest cost cell instead of the lowest item, this allocation will have associated with it a penalty corresponding to the difference of these two cost due to “Loss of advantage”. That is if we compute the difference between the two lowest costs for each row and column, we find the opportunity cost relevant to each row and column. It would be most economical to make allocations against the row or column with the highest opportunity cost. For a given row or column, the allocation should obviously be made in the least cell of that row or column.

How the method works is demonstrated by the illustration below

Example

Solve the following transportation problem using VAM

		Distribution centres					
		1	2	3	4		Supply
Plants		2	3	11	7	6	[1]
		1	0	6	1	1/0	[1]

1	5	8	15	9	10	[3]
2	7	5	3	2/1		
3						
Demand	[1]	[3]	[5]	[6]		

Steps

1. Write down the cost matrix as shown below

Enter the difference between the smallest and second smallest element in each column below the corresponding column and the difference between the smallest and second smallest element in each row to the right of the row. Put these numbers in brackets as shown.

A row or column „difference“ can be thought of a penalty for making allocations in the second smallest cost cell instead of the smallest cost cell. In other words this difference indicates the unit penalty incurred by failing to make an allocation to the smallest cost cell in that row or column in case the smallest and second smallest elements in a row/column are equal, the penalty should be taken as Zero.

2. Select the row or column with the greatest difference and allocate as much as possible within the restrictions of the rim conditions to the lowest cost cell in the row or column selected.

In case of a tie among the highest penalties, select the row or column having minimum cost in case of tie in the minimum cost also select the cell which can have maximum allocation. If there is a tie among maximum allocation cells also, select the cell arbitrarily for allocation. Following these rules yields the best possible initial basic feasible solution and reduces the number of iterations required to reach the optimal solution.

Thus, since [6] is the greatest number in brackets, we choose column 4 and allocate as much as possible to cell [2,4] as it has the lowest cost 1 in column 4. Since supply is 1 while the requirement is 2, maximum possible allocation is (1)

3. Cross out the row or column completely satisfied by the allocation just made. For the assignment just made at (2, 4), supply of plant 2 is completely satisfied. So, row 2 is crossed out and the shrunken matrix is written below.
-

1 2 3 4

1	2	3	(5)	11	7	6/1	[1]
2	5	8		15	9	10	[3]
7		5/0		3	2/1		
	[3]	[5]		[4]	[2]		

This matrix consists of the rows and columns where allocations have not yet been made including revised row and column totals which take the already made allocations into account.

4. Repeat steps 1 to 3 until all assignments have been made.

(a) Column 2 exhibits the greatest difference of [5]. Therefore we allocate (5) units to cell (1, 2), since it has the smallest transportation cost in column 2. Since requirements of column 2 are completely satisfied, this column is crossed out and the reduced matrix is rewritten below.

1 3 4

1	2	11	7	6/1	[5]
3	5	15	9	10	[4]
7		3	2/1		
	[3]	[4]	[2]		

(b) Differences are recalculated. The maximum difference is [5]. Therefore we allocate (1) to the cell (1,1) since. It has the lowest cost in row 1. Since

requirements of row 1 are fully satisfied, it is crossed out and the reduced matrix is written below.

	1	3	4	
3	5	15	9	10/4/3/0
	(6)	(3)	(1)	
	6/0	3/0	1/0	

In table above it is possible to find row different but not the column difference. Therefore, allocations in this table are made using least cost method. As cell (3, 1) has the lowest cost method.

- (c) As cell (3,1) has the lowest cost 5, max possible allocation of (6) is made here. Likewise, next allocation of (1) is made in cell (3,4) and (30 in cell (3,3) as shown.

All allocations made during the above procedure are shown in the allocation matrix

Distribution Centers

		1	2	3	4	
plants	1	2	3	11	7	6/1/0
		(1)	5			
	2	1	0	6	1	1/0
					(1)	
	3	5	8	15	9	10/4/3/0
		(6)		(3)	(1)	

$$7 \quad \frac{5}{3} \quad \frac{2}{1}$$

The cost of transportation associated with the above solution is

$$Z = (2 \times 1 + 3 \times 5 + 1 \times 1 + 5 \times 6 + 15 \times 3 + 9 \times 1)$$

$$= \underline{\underline{\pounds 10,200}}$$

The Assignment models

Given „n“ facilities and „n“ jobs and given the effectiveness of each facility for each job, the problem is to assign each facility to one and only one job so as to optimize the given measure of effectiveness. We demonstrate the procedure by examining the illustrations below:

Example 1

A machine tool company decides to make four subassemblies through 4 contractors. Each contractor is to receive only one sub assembly. The cost of each sub assembly is determined by the bids submitted by each contractor and is shown below in hundred £.

		Contractor			
		1	2	3	4
Sub assembly	1	15	13	14	17
	2	11	12	15	13
	3	13	12	10	11
	4	15	17	14	16

Assign the different subassemblies to contractors so as to minimize total cost

Solution

Step 1; We reduce the matrix and the steps are:

1. In the effectiveness matrix, subtract the minimum element of each row from all elements of that row. The resulting matrix will have at least one zero element in each row. Check if there is at least one zero element in each column also.
 2. Mark columns that do not have zero elements. Now subtract the minimum element of each of such columns from all elements of that column.
-

Contractors

	1	2	3	4
1	2	0	1	4
2	0	1	4	2
3	3	2	0	1
4	1	3	0	2

Column 4 has no zero elements. We go to sub- step 2 and subtract the minimum element 1 from all its elements.

	1	2	3	4	
Sub assembly	1	2	0	1	3
	2	0	1	4	1
	3	3	2	0	0
	4	1	3	0	1

Step 2: check if optimal assignment can be made in the current solution or not.

- Basis for this check is that check if the minimum number of lines crossing all zeros is less than n (in our example $n=4$), then an optimal assignment cannot be made in the current solution.
- If it is equal to n ($=4$) then an optimal assignment can be made in the current solution.
- Approach for obtaining minimum number of lines crossing all zeros consist of the following sub steps:
 - (a) Examine rows successively until a row with exactly one unmarked zero is found. Make a square. Around this zero, indicating that an assignment will be made there. Mark (x) all other zeros in the same column showing that they cannot be used for making other assignments
- Row 1 has single unmarked zero in column 2. Make an assignment there enclosing that zero. It means subassembly 1 is assigned to Contractor 2. Since Contractor 2 has been assigned sub-assembly 1 and as a Contractor can be assigned only one sub

assembly, any other zero in Column 2 is crossed. Since there is no other zero in this column, crossing is not required.

- Next row 2 has a single unmarked zero in column 1, make an assignment. Row 4 has a single unmarked zero in column 3, make an assignment and cross the 2nd zero in column 3. Now, row 3 has a single unmarked zero in column 4, make an assignment.
- This is shown in matrix below.

Sub assembly		1	2	3	4
	1	2	0	1	3
	2	0	1	4	1
	3	3	2	0	0
	4	1	3	0	1

Sub step 2:

Examine columns for single unmarked zeros making them and also marking x any other zero in their rows.

In case there is no row or column containing single unmarked zero (there are more than one unmarked zeros), mark one of the unmarked zeros arbitrarily and (x) all other zeros in its row and column. Repeat the process till no unmarked zero is left in cost matrix.

Minimum total cost is

$$(13 \times 1 + 11 \times 1 + 11 \times 1 + 14 \times 1) \times 100 = \text{£}4,900$$

Optimal assignment policy

Subassembly	1	Contractor	2
	2		1
	3		4
	4		3

Example 2

Four different jobs can be done on 4 different machines. The set up and take down time costs are assumed to be prohibitively high for changeover. The matrix below

gives the cost in rupees of producing job i on machine j.

Jobs		M ₁	M ₂	M ₃	M ₄
	J ₁	5	7	11	6

J ₂	8	5	9	6
J ₃	4	7	10	7
J ₄	10	4	8	3

How should the jobs be assigned to the various machines so that the total cost is minimized?

Solution

We use Hungarian method or the reduced matrix method.

Step 1

Reduce the matrix

	M ₁	M ₂	M ₃	M ₄
J ₁	0	2	6	1
J ₂	3	0	4	1
J ₃	0	3	6	3
J ₄	7	1	5	0

→

	M ₁	M ₂	M ₃	M ₄
J ₁	0	2	2	1
J ₂	3	0	0	1
J ₃	0	3	2	3
J ₄	7	1	1	0

Check if optimal assignment can be made in the current solution or not.

1. Examine rows successively until a row with exactly one unmarked zero is found. Mark this zero, indicating that an assignment will be made there. Mark (x) all other zeros in the same column showing that they cannot be used to make other assignments. Proceed in this manner until all rows have been examined.

	M ₁	M ₂	M ₃	M ₄
J ₁	 0	2	2	1
J ₂	3	 0	0	1
J ₃	0	3	2	3
J ₄	7	1	1	 0

2. Examine columns for single unmarked zeros, marking them and also marking (x) any other zeros in their rows.

3. In the present example, after following sub steps 1 and 2, we find that their repetition is unnecessary and row 3 and column 3 are without any assignments. Hence we proceed as follows to find the minimum number of lines crossing all zeros.

	M ₁	M ₂	M ₃	M ₄	
J ₁	0	2	2	1	√
J ₂	3	0	0	1	
J ₃	0	3	2	3	
J ₄	7	1	1	0	
	√				

Mark (√) the row for which assignments have not been made. In our case it is 3rd row.

4. Mark √ columns (not already marked) which have zeros in marked rows, thus column 1 is marked (√).
5. Mark (√) rows (not already marked) which have assignments in marked columns. Thus row 1 is marked (√).
6. Repeat steps 4 and 5 until no more marking is possible.
7. Draw lines through all unmarked rows and through all marked columns. This gives the minimum number of lines crossing all zeros. If the procedure is correct, there will be as many lines as the no. of assignments. In this example, the number of such lines is 3 which is less than n (n= 4 here). Hence optimal assignment is not possible in the current solution. Further reduction is necessary.
8. Examine elements that do not have a line through them. Select the smallest of these elements and subtract it from it all elements which do not have a line through. Add this smallest element to every element that lies at the intersection of two lines. Leave the remaining elements of the matrix unchanged. We obtain this matrix.

	M ₁	M ₂	M ₃	M ₄
J ₁	0	1	1	0
J ₂	4	0	0	4
J ₃	0	2	1	2
J ₄	8	1	1	0

Since the minimum number of lines passing through all zeros is 3 (< 4) optimal assignment cannot be made in the current solution.

We therefore iterate towards optimality

	M ₁	M ₂	M ₃	M ₄	
J ₁	0	>0	<0	<0	
J ₂	5	0	>0	2	
J ₃	0	1	0	2	
J ₄	8	>0	<0	0	Optimal assignment policy

Job J₁ should be assigned to machine M₁

J₂ should be assigned to machine

M₂ J₃ should be assigned to machine

M₃ J₄ should be assigned to machine

M₄

Optimal cost = £ (5 + 5+10 +3) = £ 23

4. QUEUING THEORY

- Queues or waiting lines are common in everyday life.
- We often face the problem of long queues for a bus, a movie ticket and for various other situations.
- Long queues are generally seen in front of railway booking offices and bank counters
- Queuing models are those where a facility performs a service
- A queuing problem arises when the current service rate of a facility falls short of the current flow rate of customers.
- If the queue is large it may discourage customers who may leave the queue and if this happens, then a sale is lost by the concerned business unit.
- A queue is a line that forms in front of a service facility. A queue, thus involves arriving customers (or items) who wait to be served at the facility which provide the service.

Terms commonly used in Queuing theory

- i. Customer μ - persons or units arriving at a station for service. Customers may be either persons or machines or other items.
- ii. Service station μ - Point where service is provided.
- iii. Waiting time μ the time a customer spends in the queue before being served.
- iv. Time spend by a customer in the system μ - waiting time plus service time
- v. Number of customers in the system μ - number in the queue plus those being served
- vi. Queue length μ - number of customers waiting in the queue.
- vii. Jockeying μ - joining another queue and leaving the first one
- viii. Reneging μ - joining the queue and leaving it afterwards
- ix. Balking μ - customer decides not to join the queue.
- x. Queuing system- system consisting of arrival of customers, waiting in queue, picked up for service according to a certain discipline, being served and the departure of customers.

Objectives of queuing theory

- To minimize the total waiting and service costs.
- This is done by estimating different characteristics of the waiting line such as
 - i. Average arrival rate
 - ii. Average service rate
 - iii. Average length of the queue
 - iv. Average waiting time.
 - v. Average time spent in the system

ELEMENTS OF QUEUING SYSTEM

- i. Input process or the arrivals
 - ii. Service mechanism
 - iii. Queue discipline
 - iv. Output of the queue
-

The Input Process

- Customers (or units) arrive at a service station for service.
- They do not come at regular intervals but arrivals into the system occur according to some chance mechanism.
- Often arrivals occur at random and are independent of what has previously occurred.
- Arrivals may occur at a constant rate or may be in accordance with some probability distribution, such as normal distribution, Poisson etc.
- Concerning input process, the following information is considered important
 - i. Arrival distribution
 - ii. Inter-arrival distribution
 - iii. Mean arrival rate (or the average number of customers arriving in one unit of time). We represent the mean arrival rate by the letter λ
 - iv. Mean time between arrivals $\frac{1}{\lambda}$

Service Mechanism

- Is concerned with service time and service facilities.
- Service time can either be fixed like in ATM or distributed in accordance with some probability distribution

Types of service facilities

- i. Single channel facility
 - ii. One –queue- several service stations
 - iii. Several queues- one service station
 - iv. Multi-channel facility
 - v. Multistage channel facilities
-
- The following information is considered important concerning service mechanism
 - i. Distribution of number of customers served
 - ii. Distribution of time taken to serve a customer
 - iii. Average number of customers being served in one unit of time at a service station represented by λ .
 - iv. Average time taken to serve a customer represented by $\frac{1}{\mu}$

Queue discipline

- May refer to the order in which the service station selects the next customer from the waiting line to be served. It may be like first in first out (FIFO), last in first out (LIFO), or it may be on the basis of some other criteria, sometimes, a customer may be given priority basis for service.
 - Queue discipline may also refer to the manner in which the customers form into a queue and the manner in which they behave while in the queue
-

Output of the queue

Is concerned with departure from the system.

Queuing models

- i. Fixed arrival and fixed service time
- ii. Random arrivals

Fixed arrival and fixed service time

- When the arrival and service times are fixed and same, then no backlog occurs.
- Even when the arrival and service times are not same but the arrival and service times are fixed, then a solution is to have the facilities exactly scheduled to meet the arrivals

Random arrivals

- Random arrivals of customers may be from an infinite population or from a finite population and it may be when there is one service stations or multiple service stations

Assumptions of the simplest queuing models in case of random arrivals

- One queue and it is sufficiently long
- One service station
- Queue disciplineµ First come first servedµ this means that customers waiting in the queue do not get discouraged because of queue length and there is no balking.
- Calling population is infinite
- Arrivals and services are coming individually.
- Arrivals and services occur in accordance with a Poisson process(i.e. Arrivals and services are independent of each other and mean arrival and service rates do not change overtime
- The time intervals between arrivals(i.e. The waiting time between successive events or what is called (inter- arrival time) follow exponential distribution and as such the mean inter- arrival time is represented as $\frac{1}{\lambda}$

Formulae concerning the simplest queuing model

- 1) Expected number of units (or customers) in the waiting line or being serviced (i.e., the expected number in the system isµ

$$E(n) = \frac{\lambda}{\mu - \lambda}$$

- 2) Expected number of units or customers in the queue is given by

$$E(nq) = \frac{\lambda^2}{\mu(\mu - \lambda)}$$

- 3) Average waiting time in the queue of an arrival

$$E(w) = \frac{\lambda}{\mu(\mu - \lambda)}$$

- 4) Average time an arrival spends in the system

$$E(v) = \frac{1}{(\mu - \lambda)}$$

- 5) Probability that the number in the queue and being served is greater than k

$$p(n > k) = \left(\frac{\lambda}{\mu} \right)^{k+1}$$

- 6) The expected proportion of time a facility will be idle

$$E(\text{idle}) = 1 - \frac{\lambda}{\mu}$$

- 7) The average length of non-empty queue

$$l_n = \frac{\mu}{\mu - \lambda}$$

- 8) The probability that at a given time the exact number of customers say n in the system is

$$p(n) = \left(\frac{\lambda}{\mu} \right)^n \left(1 - \frac{\lambda}{\mu} \right)$$

- 9) Utilization factor- i.e., the proportion of time a server actually spends with the customer. It is also called traffic intensity.

$$\rho = \left(\frac{\lambda}{\mu} \right)$$

Example

At a particular petrol station, arrival and services of customers follow a Poisson process with 8 customers arriving per hour and 12 customers being served per hour.

Required

- i. What is the average number of customers waiting for service?
- ii. What is the average time a customer must wait in the queue?
- iii. What is the average time for a customer to be in the system?

Solution

- Mean arrival rate of customers (λ) = 8 per hour
- Mean service time (or μ) = 12 per hour.

i. i) $E(n_q) = \frac{\lambda^2}{\mu(\mu - \lambda)}$

$$= \frac{8^2}{12(12 - 8)}$$

$$1. = 1.33$$

- ii. Average time a customer must wait in the queue

$$E(w) = \frac{\lambda}{\mu(\mu - \lambda)}$$

$$= \frac{8}{12(12 - 8)} \\ = 10 \text{ minutes}$$

- iii. Average time for a customer to be in the system

Example two

At a certain shop counter, customer's arrival follows a Poisson process with an average time of 5 minutes between arrivals. The time interval between services at the counter follows exponential distribution and as such the mean time taken to service a unit is 2 minutes. Estimate μ

- The expected average queue length
- The average number of customers in the system
- How long on average does a customer wait in the queue?
- By how much should the flow of customers be increased to justify the opening of a Second service point if the management is willing to open the same provided the customer has to wait for 5 minutes for the service?

Solution

Mean inter arrival time = 5 minutes or $\frac{1}{\lambda}$ making λ the subject of the formula we obtain

$\lambda = 0.2$. Hence mean arrival rate per hour is $0.2 \times 60 = 12$ per hour

Similarly, mean service time (or $\frac{1}{\mu}$) = 2 minutes or $\mu = 30$ per hour

- i) Expected average queue length

$$E(nq) = \frac{\lambda^2}{\mu(\mu - \lambda)} \\ = \frac{12^2}{30(30 - 12)} \\ = 0.267$$

ii) Average number of customers in the queuing system

$$\begin{aligned} E(n) &= \frac{\lambda}{\mu - \lambda} \\ &= \frac{12}{30 - 12} \\ &= 0.667 \end{aligned}$$

iii) Average waiting time in the queue

$$\begin{aligned} E(w) &= \frac{\lambda}{\mu(\mu - \lambda)} \\ &= \frac{12}{30(30 - 12)} \\ &= 0.022hr \end{aligned}$$

iv) $E(W) = 5$ or $\frac{1}{12}$ hrs and the corresponding mean arrival rate (or λ') for which would be worked out as follows:

$$\begin{aligned} E(W) &= \frac{\lambda'}{\mu(\mu - \lambda')} \\ &= \frac{1}{12} = \frac{\lambda'}{30(30 - \lambda')} \end{aligned}$$

Making λ' the subject of the formula we obtain

$$\lambda' = 21.43$$

Hence the mean arrival rate should go up from the present rate of 12 per hour to 21.43 per hour to justify the opening of a second service point.



Self Assessment Questions

1. Discuss the limitations in a queuing model
 2. A self service store employs one cashier at its counter. Nine customers arrive on an average every 5 minutes while the cashier can serve 10 customers in 5 minutes. Assuming Poisson distribution for arrival rate and exponential distribution for service time, find:
-

- i. Average number of customers in the system
 - ii. Average queue length
 - iii. Average time a customer spends in the system
 - iv. Average time a customer waits before being served
 3. A person repairing radios finds that the time spent on the radio sets has exponential distribution with mean 20 minutes. If the radios are repaired in the order in which they come in and their arrivals is approximately Poisson with an average rate of 15 minutes for 8- hour day, what is the repairman's expected idle time each day? How many jobs are ahead of the average set just brought in?
 4. Arrival rate of telephone calls at a telephone booth are according to Poisson distribution, with an average time of 9 minutes between two consecutive arrivals. The length of telephone call is assumed to be exponentially distributed with mean 3 minutes.
 - i. Determine the probability that a person arriving at the booth will have to wait.
 - ii. The telephony company will install a second booth when convinced that an arrival would expect to have to wait at least four minutes for the phone. Find the increase in the flow rate of arrivals which will justify a second booth
 - iii. Find the fraction of the day that the phone will be in use
 - iv. Find the average queue length that is formed from time to time.
 5. Workers come to tool store room to receive special tools (required by them) for accomplishing a particular project assigned to them. The average time between two arrivals is 60 seconds and the arrivals are assumed to be in Poisson distribution. The average service time (of the tool room attendant) is 40 seconds. Determine:
 - i. Average queue length
 - ii. Average length of non empty queue
 - iii. Average no of workers in the system including the workers being attended
 - iv. The mean waiting time of an arrival.
 6. A branch of ABC bank has only one typist. Since the typing work varies in length (number of pages to be typed), the typing rate is randomly distributed approximating a Poisson distribution with mean service rate of 8 letters per hour. The letters arrive at a rate of 5 per hour during the entire 8- hours work day. If the type writer is valued at £2 per hour, determine:
 - i. Equipment utilization
 - ii. The percent time that an arriving letter has to wait
 - iii. Average system time
 - iv. Average cost due to waiting on the part of the type writer.
-

Further reading



Read on:

- i. Erlang's Method of solving queuing problems.
- ii. Queuing models in case of multiple service station

5. NETWORK ANALYSIS IN PROJECT PLANNING (PERT & CPM)

Introduction

Project is defined as a combination of interrelated activities which must be executed in a certain order before the entire task is completed.

Characteristics of a project

- A project has identifiable beginning and end points; it is an entity by itself.
- It is not a permanent entity, it is usually a non-repetitive task.
- It can be broken down into identifiable activities which require time and resource for their execution.
- It is scheduled to be completed by a target date.
- The objectives are clear and output or end product definite
- It is usually large and complex with time horizon of 2 t 3 years. However, some projects have taken more than 10 years while others have lasted for less than 6 months.
- It usually involves heavy investment.
- Execution of projects activities and hence completion of the project is always subject to some uncertainties and risks.
- Each project, whether small or big, has three basic requirements.
 - i) It should be completed without any delay.
 - ii) It should use available manpower and other resources economically.
 - iii) It should involve as small investment as possible.
- The project management helps to fulfil the above requirements
- The project management involves the following three phases
 - i) Project planning
 - ii) Project scheduling
 - iii) Project controlling.

Out of these three phases, the first two are accomplished before the start of the actual project. The 3rd phase comes into operation during the execution of the project. It highlights the bottlenecks during execution of the project and spells out measures to remove bottlenecks & complete the project in time.

The network technique of PERT & cpm are valuable during all the three phases of project management.

1. PROJECT PLANNING

Is an important phase during which are set in plans and strategies of project execution, keeping in mind the policies, procedures and rules of the organization. It has two important aspects.

- i) Identification of the activities (jobs or tasks) to be performed and,
- ii) Estimation of the required resources.

Resources are of many kinds and are classified as:

- i) Manpower resources
- ii) Equipment (machinery) resources
- iii) Financial resources
- iv) Material resources
- v) Time and space resources.

Since resources are the starting points of projects, their proper planning and estimation helps to estimate the time and cost of the various activities and hence of the entire project.

Steps in project planning

- i) Setting the objectives of the project and the assumptions to be made.
 - ii) Developing the W.B.S (Work Breakdown Structure). Depending upon the objective of the management, the extent of control desired and the availability of computational aids, the project is broken down into clearly definable activities.
 - iii) Determination of time estimate of these activities
 - iv) Estimation of resources –financial, managerial and operational.
 - v) Study of alternative ways of achieving the goals.
 - vi) Establishment of inter-dependence relationship between the activities i.e. the sequence of performing the activities.
-

2. PROJECT SCHEDULING

Scheduling is the laying out of the project activities along a time sequence in which they are to be performed so as to assign the starting and finishing dates to various activities and to allocate resources to them. In other words, scheduling is the preparation of timetable of the implementation of the activities and computation of resources required at different stages of time.

It also includes identifying the tasks that are critical and the resources that are limited so that the entire project proceeds in a logical, orderly and systematic manner. In short, the scheduling phase consists of determining.

- i) Start and finish times of each activity; the earliest and latest times at which events can occur.
- ii) Critical activities that require special attention
- iii) Allocating resources – men, machines, materials, etc. to each activity.
- iv) Slacks and floats for non-critical activities
- v) Various constraints due to limitation of resources.

3. PROJECT CONTROLLING

Controlling phase is the follow up of the planning and scheduling phases.

While planning and scheduling are undertaken before actual project starts, the controlling phase is undertaken during the actual execution of the project.

The project is monitored to find deviations in actual progress from the scheduled plan and to apply the corrective measure, so as to achieve the targets.

Project controlling usually involves the following steps.

- i) Settling standard and targets with regard to time and cost of the project
 - ii) Reviewing the progress by comparing the work accomplished to work scheduled at different stages of time and finding the deviations.
 - iii) Evaluating the effect of deviations on the project plan.
 - iv) Updating the project schedule
 - v) The corrective measures to rectify the deviations from the plan should be suggested. This requires decision making with regard to scheduling of resources, scheduling of jobs, crashing of projects. Etc.
-

WORK BREAKDOWN STRUCTURE (W.B.S)

A project is a combination of interrelated activities which must be performed in a certain order for its completion. The process of dividing the project into these activities is called Work Breakdown Structure (W.B.S). The activity or unit of work or work content is a clearly identifiable and manageable work unit. The number of activities or the size of each activity will depend upon the level of detail desired.

BASIC TOOLS AND TECHNIQUES OF PROJECT MANAGEMENT

The various tools and techniques of project management are grouped into the following two heads.

- i) Bar charts, milestone charts, and velocity diagrams.
- ii) Network techniques

Bar Charts, milestone charts and velocity diagrams.

Bar charts are the two-dimensional pictorial representation of a project. In a bar chart, the activities of the project are shown on one axis and their duration are represented on the other axis. A bar chart helps to review the project progress, allows for rescheduling the project and highlights the critical activities and the bottlenecks in the completion of the project. A bar chart, however, is normally suited to small projects. It cannot take into account the uncertainties in activity duration nor represent the interrelationships between the various activities of the project.

Milestone charts are the modified and improved versions of bar charts. However, whereas bar chart represents activities, milestone charts represent the events which mark either the beginning or the end of an activity. Though milestone charts are

more detailed and offer better control than bar charts, they possess most of the drawbacks inherent in bar charts.

Velocity diagrams are useful for representing the activities which require a series of crews working in a given sequence.

Network Techniques

A network (also called network diagram or network technique) is a symbolic representation of the essential characteristics of a project. PERT & CPM are the two most widely applied techniques of network analysis.

1. Programme Evaluation and Review technique (PERT)

It uses event oriented network in which successive events are joined by arrows. It is preferred for projects that are non-repetitive and in which time for various activities cannot be precisely pre-determined.

There is no significant past experience to guide; they are once-through projects.

Launching a new product in the market by a company, research and development of a new war weapon, launching of satellite, sending space craft to mars are examples of PERT projects.

Three time estimates – the optimistic time estimate, pessimistic time estimate and the most likely time estimates are associated with each and every activity to take into account the uncertainty in their times. These estimates are explained below:

Pessimistic time

In Program Evaluation and Review Techniques (PERT) it is the longest time in which an activity is likely to be completed.

Optimistic time

It is the shortest possible time in which an activity is likely to be completed.

Most likely time.

The time that the activity would most likely take if it were repeated time and time again.

2. Critical Path Method (CPM)

- It uses activity oriented network which consists of a number of well recognized jobs, tasks or activities.
- Each activity is represented by an arrow and the activities are joined together by events.
- CPM is generally used for simple repetitive types of projects for which the activity times and costs are certainly and precisely known.
- Projects such as construction of a building, road, bridge, physical verification of store, yearly closing of accounts by a company can be handled by CPM.
- Thus, it is deterministic rather than probabilistic model.

Role of network Techniques in Project management

- They help managers in planning, scheduling and controlling of projects.

NETWORK ANALYSIS

Network analysis is the organized application of systematic reasoning for planning, scheduling and controlling practical situations where many separate tasks (which collectively make up the whole project.) can happen simultaneously or consecutively such that it is difficult to establish the relationship between the separate jobs.

The technique can be applied to any purposeful chain of events involving the use of time, labour and physical resources although it is related to large industrial or commercial projects of a complex nature where the scale of operation gives rise to correspondingly greater financial administrative problems.

Network analysis involves the following three basic steps:

- i) It defines the job to be done
- ii) It integrates the various activities in a logical sequence
- iii) It controls the progress of project plan

Main techniques of network analysis are as under:

- Critical Path Analysis (CPA)
- Project Evaluation and Review Technique (PERT)

Both the techniques have been successfully applied to improve efficiency of execution of large projects within pre-determined time and costs limits.

Some of the more common applications of network analysis occurs in projects scheduling for:

- The construction of buildings, bridge, factories. Etc.
- Budgeting and auditing procedures.
- Missile development
- The installation of a computer system.
- Launching a new project
- Strategic and tactical military planning

NB: Network analysis is likely to be of most value where projects are:

- a) Complex, ie. They contain many related and interdependent activities.
- b) Large i.e. where many types of facilities, high capital investment, many personnel are involved.
- c) Where restrictions exist i.e. where projects have to be completed within stipulated time or cost limits, or where some or all of the resources (material, labour) are limited.

BASIC TERMINOLOGIES

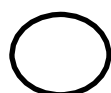
1. Activity:

This is a task or job of work which takes time and resources and has an end and beginning e.g. verify the debtors in a sales ledger, dig foundations e.t.c. In a network, activity is represented by an arrow 

The head of the arrow indicates where the task ends and the tail where it begins. The arrow points from left to right but is not drawn to scale. When drawing networks, always establish:

- a) What activities are involved in the project?
- b) Their logical relationship e.g. the activity of building the wall must take place after the activity, dig foundations.
- c) An estimate of the time an activity is expected to take.

2. **Event** ; this is point in time and indicates the start or finish of an activity, or activities, e.g. all build, debtors verified, foundations dug etc. An event in a network is represented by a circle or node as under:



3. Dummy Activities

This is an activity that does not consume time or resources. It is represented in a network by a dotted arrow➔➡

It is used merely to show clear, logical dependencies between activities so as not to violate the rules for drawing networks

NB:

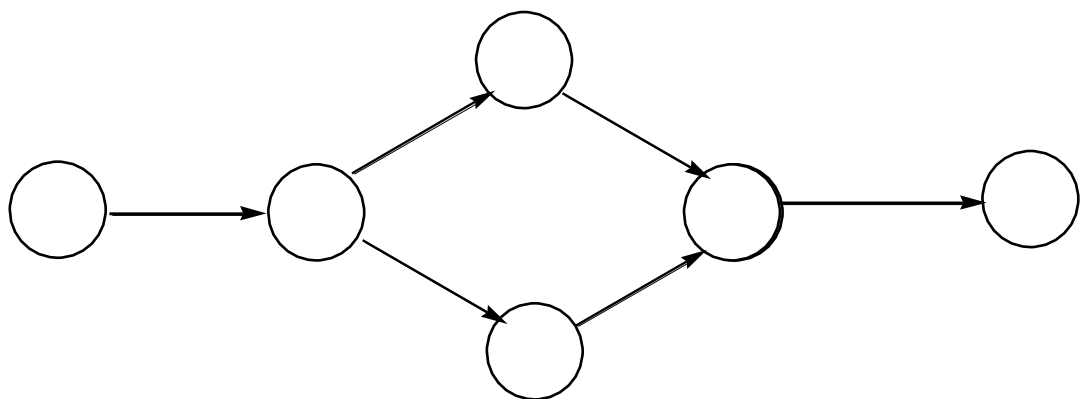
A dummy activity is created for the following reasons:

- a) To make activities with common starting and finishing events distinguishable.
- b) To identify and maintain the proper precedence between activities that are not connected by events.

4. Network

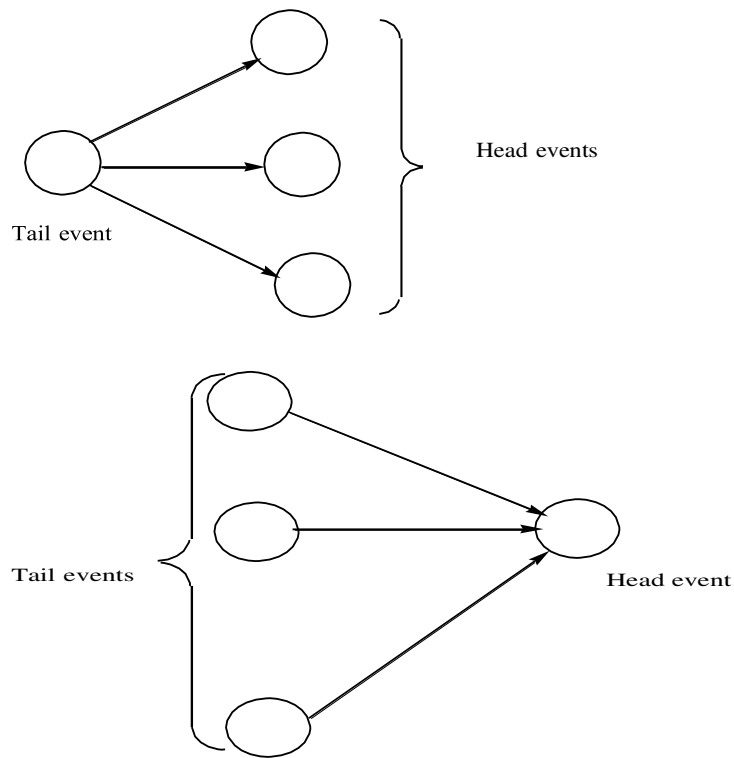
This is the combination of activities, dummy activities and events in a logical sequence according the rules for drawing networks.

It is a graphic representation of a projects operation and is composed to activities and events (nodes). In numbers i.e. 1, 2, 3, 4, -----while the letters A, B, C, D----- represent the activities a network may appear as this.



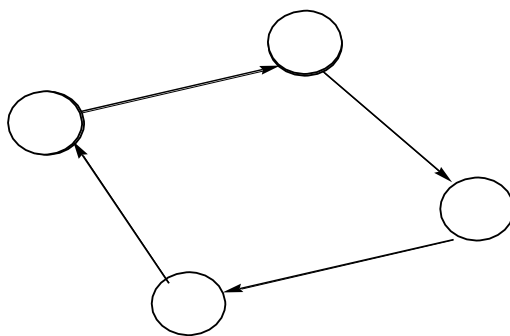
Rules for drawing networks

- i) A complete network should have only one point of entry a start event and only one point of exit – a finish event
- ii) Every activity must have one preceding or “tail” event and one succeeding or “head” event. NB: many activities may use the same tail event and many may use the same head event e.g.

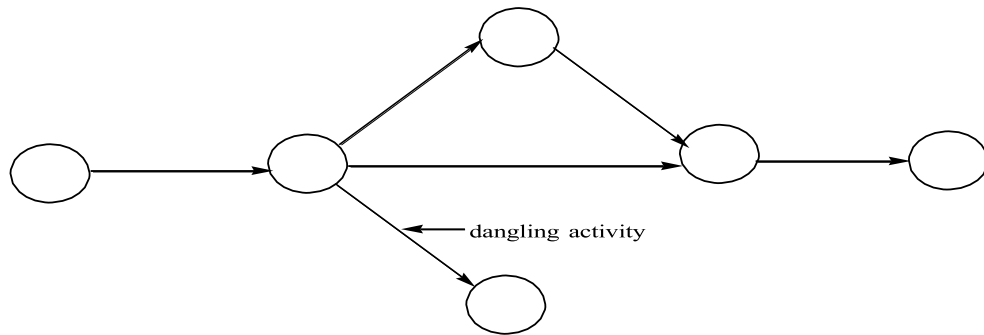


However, an activity must not share the same tail event and the same head event with any other activities.

- iii) No activity can start until all activities leading to it are complete.
- iv) “Loops i.e. a series of activities which lead back to the same event, are not allowed because the essence of network is a progression of activities always moving onwards in time.



- v) All activities must be tied into the network i.e. they must contribute to the progression or be discarded as irrelevant. Activities which do not link into the overall project are termed „danglers“



Note the following when drawing networks:

1. Networks proceed from left to right
2. Networks are not drawn to scale i.e. the length of the arrow does not represent time elapsed.
3. Arrows need not be drawn in the horizontal plane but unless it is totally unavoidable they should proceed from left to right.
4. Events or nodes should be progressively numbered from left to right. Simple networks may have events numbered in simple numeric progression. 0, 1, 2, 3 etc but larger more realistic networks may be numbered in fives i.e. 0, 5, 10, 15 etc. This enables additional activities to be inserted subsequently without affecting the numbering sequence of the whole project.

NB: there is one starting event and one finishing event

- Enter first the activities with no pre-requisite
- Next enter the activities which follow from the activities already in the network.

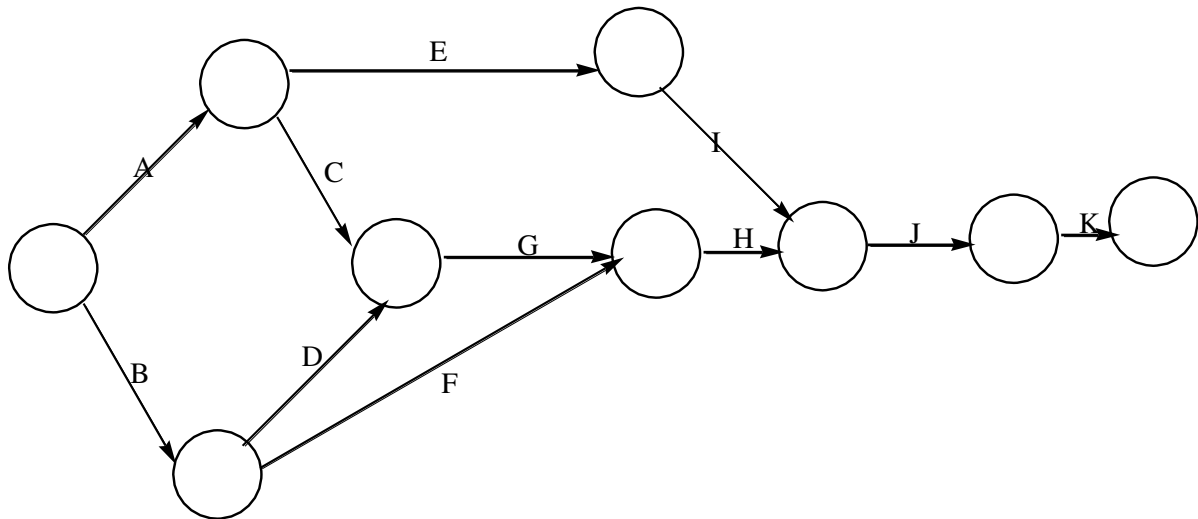
Example

Draw a network for the simple project of erection of steel works for a shed.
The various activities of the project are as under:

Activity	Description	Preceded by:
A	Erect site workshop	-
B	Fence site	-
C	Bend reinforcement	A
D	Dig foundation	B
E	Fabricate steel work	A
F	Install concrete pillars	B
G	Place reinforcement	C,D
H	concrete foundation	G,F

I	Erect steel work	E
J	Paint steel work	H,I
K	Give finishing touch	J

Solution



More definitions

1). Sub critical Activity

It is an activity having next higher float than the critical activity. It demands normal attention but allows some freedom of action. The path connecting such activities is named as the Sub- Critical path. A network may have more than one sub-critical path

2. Super Critical Activity

An activity having negative float is called supercritical activity. Such activity demands very special attention and action. It results when activity duration is more than the time available.

3. Slack

It is the time by which occurrence of an event can be delayed. It is denoted by S and is the difference between the latest occurrence times of the event.

i.e. $S = L - E$ of the event.

Time estimates



After drawing the network, it is necessary to insert the activity duration. We can have two types of time estimates

- i. Single time estimates for each activity. These estimates would be based on judgment or experience of the individual responsible or by technical calculations using data from similar projects
- ii. Multiple time estimates for each activity. The most usual multiple time estimates are three estimates for each activity. i.e. optimistic (O), most likely (M,L) and pessimistic (P). These three estimates are combined to give an expected time and the accepted formula is:

$$\text{Expected time} = \frac{O + P + 4ML}{6}$$

e.g. Assume the three estimates for activities are:

Activity	Optimistic	Pessimistic	Most likely	expected time
A	6	10	8	8
B	4	8	6	6
C	7	11	9	9
D	9	13	11	11
E	5	9	7	7
F	3	7	5	5

Note: There is no fundamental difference between the two methods as regards the basic time analysis of a network. However, on completion of basic time analysis, projects with multiple time estimates can be further analyzed to give an estimate of the probability of completing the project by a scheduled date. We shall demonstrate this towards the end of the chapter.

Time Units

Time estimates may be given in any unit, i.e. minutes, hours, days, and weeks depending on the project. All time estimates within a project must be in the same units otherwise confusion is bound to occur.

Critical Path

It is the chain of activities with the longest duration times. Critical path gives the expected project duration

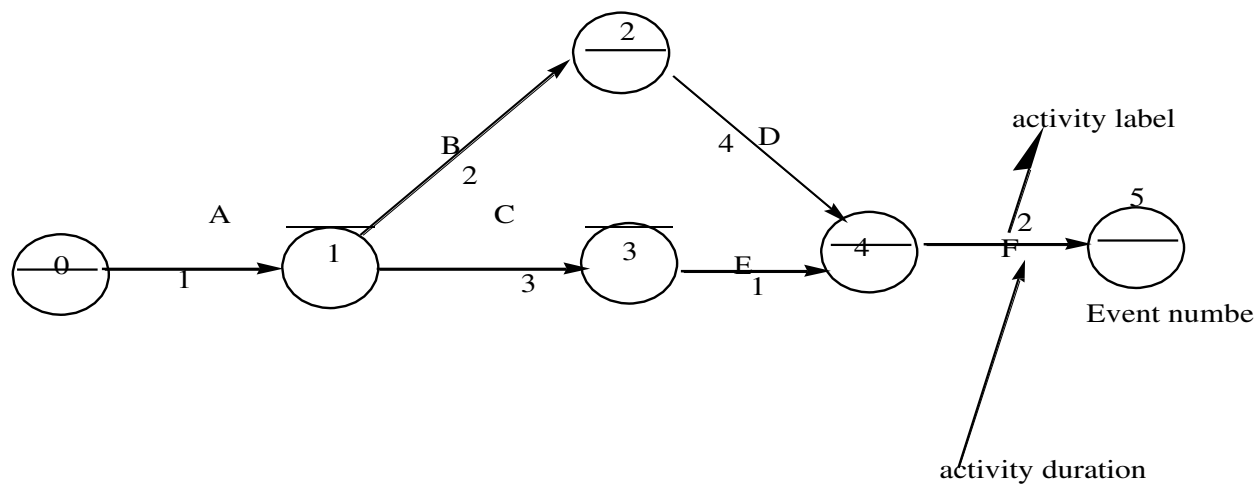
There may be more than one critical path in a network and is possible for critical path to run through a dummy.

Procedure for establishing the critical path

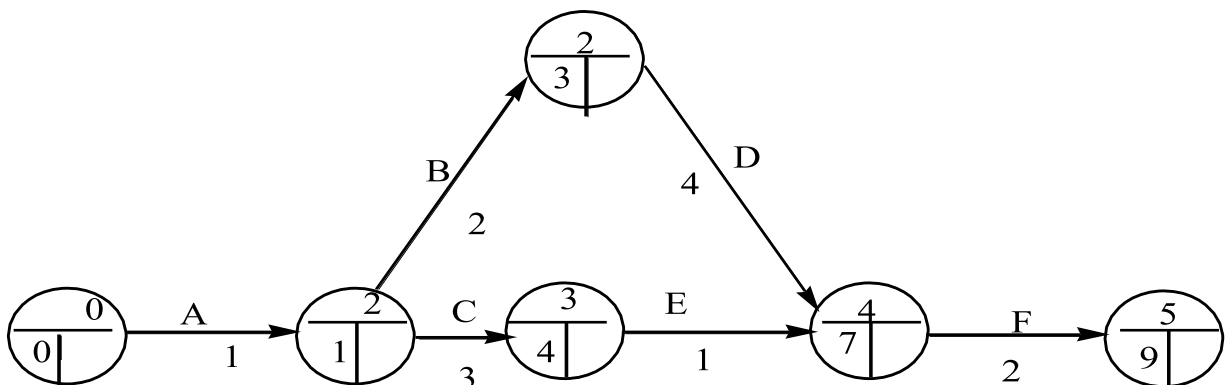
1. Earliest Start time (EST) (forward Pass)

The EST is the earliest possible time at which a succeeding activity can start e.g.

Assume the network below has been drawn and the activity time estimated in days.



The EST_s can be inserted as follows:



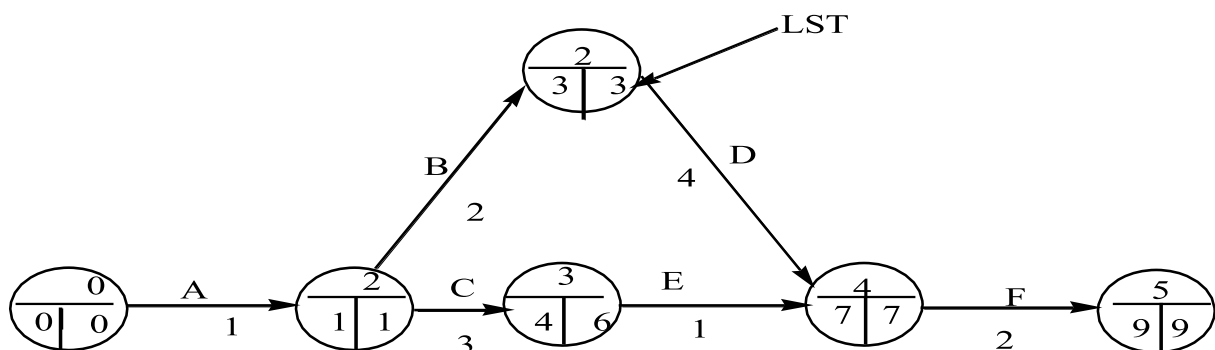
NB:

1. The EST of a head event is obtained by adding onto the EST of the tail event the linking activity duration starting from event 0, time 0 and working forward through the network.
2. Where two or more routes arrive at an event, the longest route time must be taken, e.g. activity F depends on completion of D and E. E is completed by day 5 and D is not complete until day 7. Therefore, F cannot start before day 7.
3. The EST in the finish event No. 5 is the project duration and is the shortest time in which the whole project can be completed.

2. Latest Start times (LST)

The LST is the latest possible time at which a preceding activity can finish without increasing the project duration.

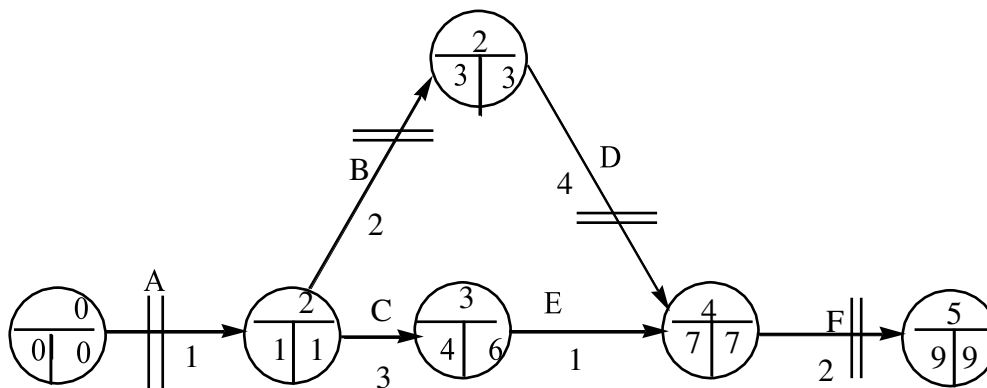
Example



Notes on calculating LST (termed the backward pass)

- a) Starting at the finish event No. 5, insert the LST (i.e. day 9) and work backwards through the network deducting each activity duration from the previously calculated LST.
 - b) Where the tails of activities B and C join event No.1, the LST for C is day 3 and the LST for B is day 1. The lowest number is taken as the LST for event NO.1 because if event No. 1 occurred at day 3 then activities B and D could not be completed by day 7 as required.
-

From example 1 Critical path can be indicated either by a different colour or by two transverse lines across the arrows on the critical path



NB. The critical path are vital activities which must be completed by their EST_s / LST_s otherwise the project will be delayed. Non-critical activities (C and E) have spare time or float available C and / or E could take up to an additional 2 days in total without delaying the project duration than the time of one or more of the activities on the critical path must be reduced perhaps by using more labour, or more better equipment or some other methods of reducing job times.

Float

Float or spare time can only be associated with activities which are non-critical. By definition, activities on the critical path cannot have float

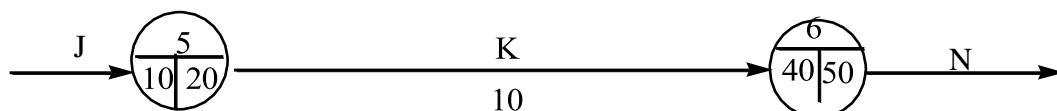
Types of float

1.) Total Float

The amount of time a path of activities could be delayed without affected the overall project duration.

Total Float = latest head time - Earliest tail time - Activity duration

Consider this part of a network.



Total float = 50 - 10 - 10

= 30 days.

2.) Free Float

This is the amount of time an activity can be delayed without affecting the commencement of a subsequent activity at its earliest start time , but may affect float of a previous activity.

Free float = Earliest Head time - Earliest tail time - Activity duration

Free float = 40- 10-10
= 20 days.

3.) Independent float

This is the amount of time an activity can be delayed when all preceding activities are completed as late as possible and all succeeding activities completed as early as possible.

It does not affect the float of either preceding or subsequent activities.

Independent float = Earliest head time - latest tail time- Activity duration

Independent float = 40 -20-10
= 10 days

4.) Interfering float= total float- free float

Slack

This is the difference between the EST and LST for each event. Events on the critical path have a zero slack.

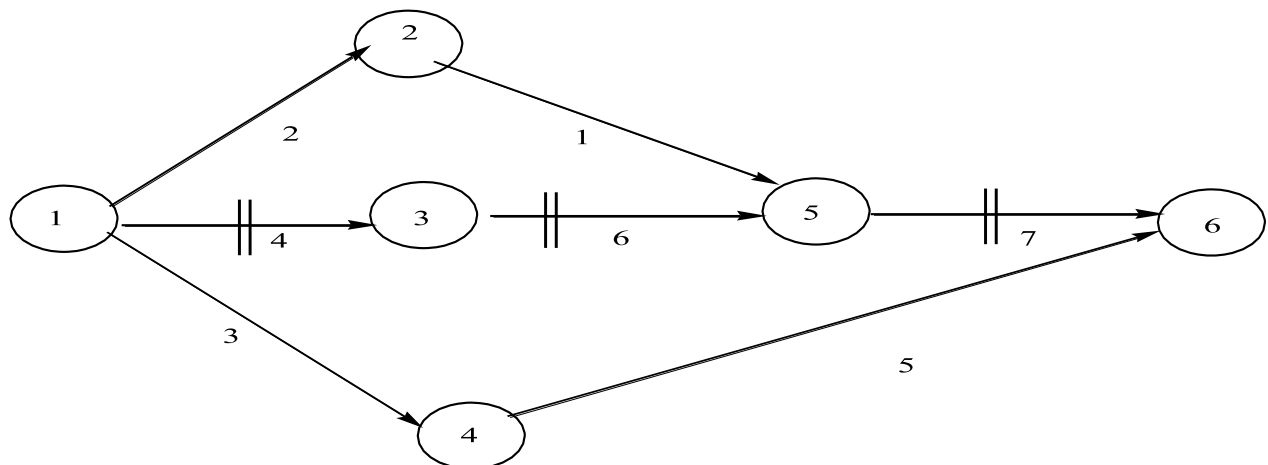
Probability

The three time estimates can be used to make statements about the probability of completing the project within a scheduled date. We are going to demonstrate this by working on the example below. The time estimates (in weeks) for the activities of a PERT network are given below:

Activity	t _o	t _m	t _p	t _e	σ ²
1 - 2	1	1	7	2	1
1 - 3	1	4	7	4	1
1 - 4	2	2	8	3	1
2 - 5	1	1	1	1	0
3 - 5	2	5	14	6	4
4 - 6	2	5	8	5	1
5 - 6	3	6	15	7	4

1. Draw the project and identify all paths through it.
 2. Determine the expected project length
-

3. Calculate the standard deviation and variance of the project length.
4. What is the probability that the project will be completed at least 4 weeks earlier than expected?
5. The probability that the project will be completed on schedule if the scheduled completion time is 20 weeks.



Critical Path 1 - 3 - 5 - 6 = 17

Variance of the project length is the sum of the variances of the activities on the critical path

$$V_{CP} = V_{1-3} + V_{3-5} + V_{5-6}$$

$$= 1+4+4 = 9$$

$$\sigma = 3 \text{ Weeks}$$

- d) Probability that the project will be completed at least 4 weeks earlier than expected time:

Expected time = 17 weeks

Scheduled time = 17-4 13 weeks

The Standard normal deviate

$$Z = \frac{13-17}{3}$$

$$= \frac{-4}{3}$$

$$= -1.33$$

For $z = -1.33$, the probability is 0.0918